# Reflection, rotation and translation on a grid

Three ways of moving a shape that leave every length and every angle alone. [Cambridge insert]

LESSON 173–174 | 2 × 40 MIN | MATEMATIKA 6, QO'SHIMCHA MAVZU | S7 14 POSITION AND TRANSFORMATION

**LESSON CLOCK**

16' | 22' | 24' | 18'

| | |
|---|---|
| The three transformations | 16 min |
| Reflection | 22 min |
| Rotation | 24 min |
| Translation | 18 min |

**LEARNING OBJECTIVES**

- Reflect a shape in the axes and in the line $y = x$.
- Rotate a shape about the origin through $90°$ and $180°$.
- Translate a shape by a given pair of numbers.
- State what each transformation keeps and what it changes.

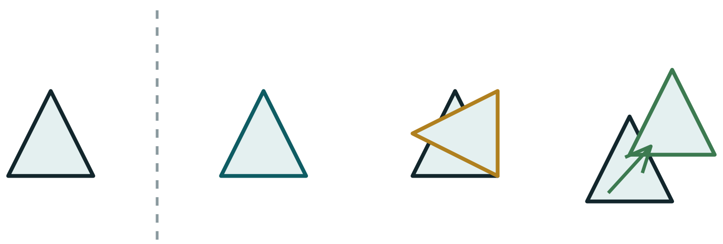
**TEXTBOOK**

| | |
|---|---|
| National | pp. 493–497 |
| Cambridge | Stage 7, pp. 144–152 |
| Workbook | Exercise 14.1–14.3 |

## 01 Explanation

### The three transformations

Each of the three moves a shape without bending or resizing it: every length and every angle of the **image** matches the original. Only the position — and for a reflection the sense — is different.




*Reflection, rotation and translation of the same triangle*

| TRANSFORMATION | NEEDS | LENGTHS | ORIENTATION |
|---|---|---|---|
| reflection | a mirror line | unchanged | **reversed** |
| rotation | a centre, an angle and a direction | unchanged | kept |
| translation | two numbers — across and up | unchanged | kept |

DESCRIBING A TRANSFORMATION FULLY IS PART OF THE ANSWER

“A rotation” is incomplete; “a rotation of $90°$ anticlockwise about the origin” is not. A reflection needs its mirror line, a translation its two numbers.

## Reflection

Each point moves straight across the mirror line to the same distance on the other side. A point already on the line does not move at all.

| MIRROR | RULE | $A(1, 1)$ | $B(4, 1)$ | $C(2, 3)$ |
|---|---|---|---|---|
| the $y$-axis | $(x, y) \rightarrow (-x, y)$ | $(-1, 1)$ | $(-4, 1)$ | $(-2, 3)$ |
| the $x$-axis | $(x, y) \rightarrow (x, -y)$ | $(1, -1)$ | $(4, -1)$ | $(2, -3)$ |
| the line $y = x$ | $(x, y) \rightarrow (y, x)$ | $(1, 1)$ | $(1, 4)$ | $(3, 2)$ |

A REFLECTION TURNS THE SHAPE OVER

Label the vertices $A$, $B$, $C$ and read them round the triangle: if they ran clockwise before, they run anticlockwise after. Nothing else in this topic does that.

## Rotation

Every point swings round the centre through the same angle, staying the same distance from it. About the origin the three useful cases have simple rules.

| ROTATION ABOUT $O$ | RULE | $A(1, 1)$ | $B(4, 1)$ | $C(2, 3)$ |
|---|---|---|---|---|
| $90°$ | anticlockwise: $(x, y) \to (-y, x)$ | $(-1, 1)$ | $(-1, 4)$ | $(-3, 2)$ |
| $180°$ | either way: $(x, y) \to (-x, -y)$ | $(-1, -1)$ | $(-4, -1)$ | $(-2, -3)$ |
| $90°$ | clockwise: $(x, y) \to (y, -x)$ | $(1, -1)$ | $(1, -4)$ | $(3, -2)$ |

THE CENTRE IS THE ONLY POINT THAT STAYS STILL

For a rotation of $180°$ the direction does not matter; for $90°$ it decides everything, so it must be stated. Tracing paper turned about a pin is the honest way to check.

## Translation

Every point slides the same distance in the same direction — so many across, so many up. Nothing turns and nothing flips.

| TRANSLATION | RULE | $A(1, 1)$ | $B(4, 1)$ | $C(2, 3)$ |
|---|---|---|---|---|
| $3$ right, $2$ down | $(x, y) \to (x + 3, y - 2)$ | $(4, -1)$ | $(7, -1)$ | $(5, 1)$ |
| $2$ left, $3$ up | $(x, y) \to (x - 2, y + 3)$ | $(-1, 4)$ | $(2, 4)$ | $(0, 6)$ |

EVERY TRANSFORMATION HERE CAN BE UNDONE

A translation of $3$ right and $2$ down is undone by $3$ left and $2$ up; a reflection is undone by itself; a rotation of $90°$ anticlockwise by $90°$ clockwise. Two reflections, in the $x$-axis then the $y$-axis, make a rotation of $180°$.

## 02 Worked examples

**EXAMPLE 1** **Reflect the triangle $A(1, 1)$, $B(4, 1)$, $C(2, 3)$ in the $y$-axis.**

1. The rule is $(x, y) \to (-x, y)$.
   Only the sign of $x$ changes.
2. $A(1, 1) \to (-1, 1)$, $B(4, 1) \to (-4, 1)$.
3. $C(2, 3) \to (-2, 3)$.
   The labels now read the other way round ✓

ANSWER

$(-1, 1)$, $(-4, 1)$ and $(-2, 3)$

**EXAMPLE 2** **Rotate the same triangle $90°$ anticlockwise about the origin.**

1. The rule is $(x, y) \to (-y, x)$.
2. $A(1, 1) \to (-1, 1)$, $B(4, 1) \to (-1, 4)$.
3. $C(2, 3) \to (-3, 2)$.
   Distances from $O$ are unchanged ✓

ANSWER

$(-1, 1)$, $(-1, 4)$ and $(-3, 2)$

EXAMPLE 3 **Translate the same triangle $3$ right and $2$ down.**

1. Add $3$ to each $x$ and subtract $2$ from each $y$.
2. $(4, -1), (7, -1)$.
3. $(5, 1)$.
   The shape has not turned at all ✓

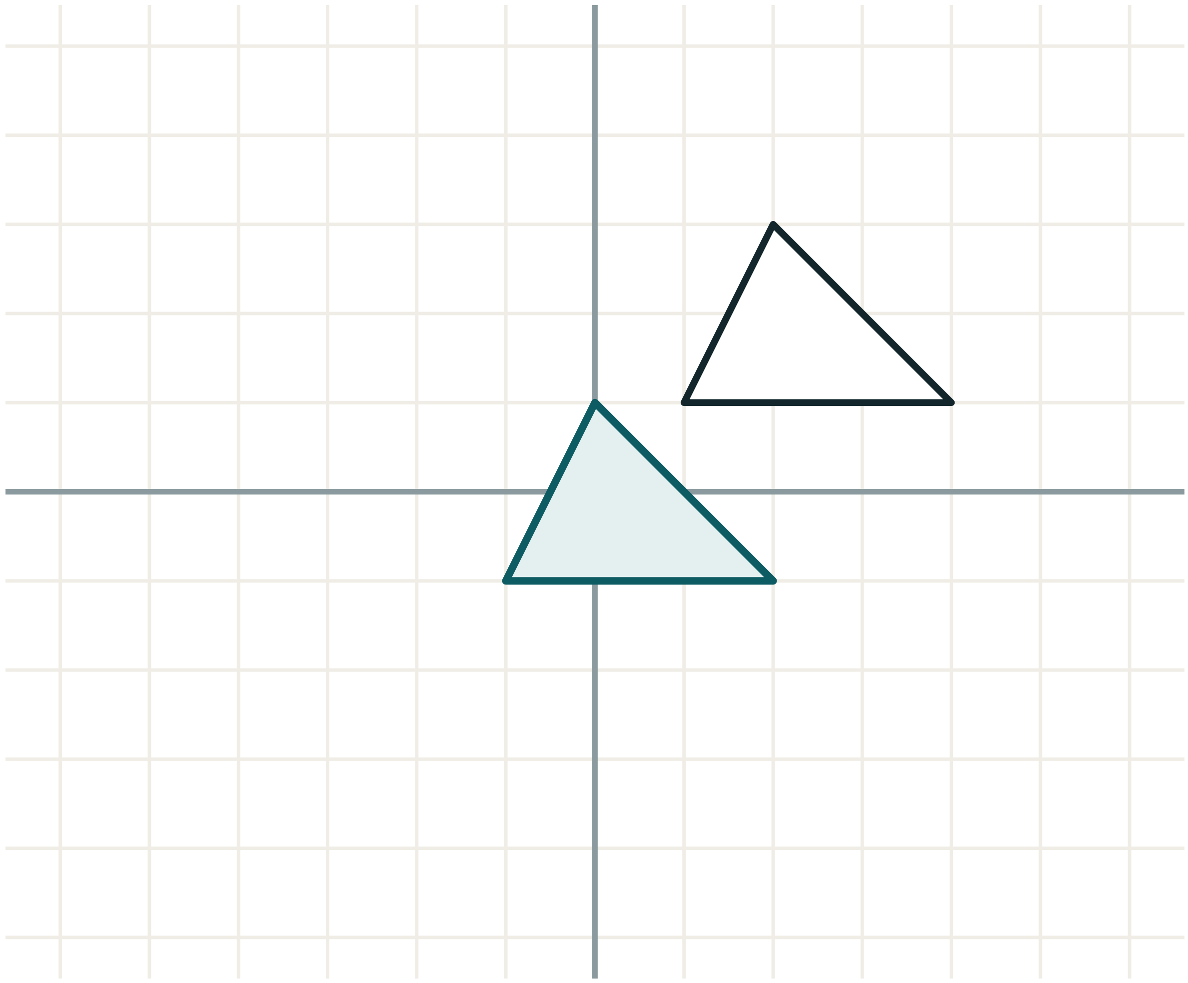
ANSWER

$(4, -1), (7, -1)$ and $(5, 1)$

## 03 Interactive model

Switch the model between reflection and rotation while watching the orientation readout; that one word is the difference the class most often misses.

### Move the shape



transformation **translation**

lengths **unchanged**

angles **unchanged**

orientation **kept**

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH | O‘ZBEKCHA | РУССКИЙ |
|---|---|---|
| Transformation | Almashtirish | Преобразование |
| Reflection | Simmetriya (aks ettirish) | Отражение |
| Mirror line | Simmetriya o‘qi | Ось симметрии |
| Rotation | Burish | Поворот |
| Centre of rotation | Burilish markazi | Центр поворота |
| Translation | Parallel ko‘chirish | Параллельный перенос |
| Image | Tasvir | Образ |
| Orientation | Yo‘nalish | Ориентация |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

## 05 Quick check

### Check yourself

Reflection in the $y$-axis sends $(3, 2)$ to:

- $(3, -2)$
- $(-3, 2)$
- $(-3, -2)$
- $(2, 3)$

A rotation of $180°$ about $O$ sends $(3, 2)$ to:

- $(-3, 2)$
- $(3, -2)$
- $(-3, -2)$
- $(2, 3)$

Which transformation reverses the orientation?

- rotation
- translation
- reflection
- none

A rotation must be described with:

- a mirror line
- a centre, an angle and a direction
- two numbers
- nothing

A translation of $3$ right and $2$ down is undone by:

- $3$ right, $2$ up
- $3$ left, $2$ up
- $2$ left, $3$ up
- itself

Under all three transformations the lengths are:

- doubled
- halved
- unchanged
- reversed

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

### Easy · warm the idea up 7 PROBLEMS

1. Reflection in the $y$-axis sends $(3, 2)$ to

   ANSWER $(-3, 2)$

2. Reflection in the $x$-axis sends $(3, 2)$ to

   ANSWER $(3, -2)$

3. A translation of $2$ right and $3$ up sends $(1, 1)$ to

   ANSWER $(3, 4)$

4. A rotation of $180°$ about $O$ sends $(3, 2)$ to

   ANSWER $(-3, -2)$

5. All three transformations keep

   ANSWER The lengths and the angles

6. A reflection needs

   ANSWER A mirror line

7. A rotation needs

   ANSWER A centre, an angle and a direction

## Medium · the standard question

7 PROBLEMS

1. Reflection in $y = x$ sends $(4, 1)$ to

   ANSWER $(1, 4)$

2. A $90°$ anticlockwise rotation about $O$ sends $(4, 1)$ to

   ANSWER $(-1, 4)$

3. A $90°$ clockwise rotation about $O$ sends $(4, 1)$ to

   ANSWER $(1, -4)$

4. A translation of $3$ right and $2$ down sends $(2, 3)$ to

   ANSWER $(5, 1)$

5. Reflect $A(1, 1)$, $B(4, 1)$, $C(2, 3)$ in the $y$-axis

   ANSWER $(-1, 1)$, $(-4, 1)$, $(-2, 3)$

6. Which transformation reverses the labels round the shape?

   ANSWER Reflection

7. A translation is described by

   ANSWER Two numbers — across and up

## Hard · stretch and reason

7 PROBLEMS

1. Rotate $A(1, 1)$, $B(4, 1)$, $C(2, 3)$ by $180°$ about $O$

   ANSWER $(-1, -1)$, $(-4, -1)$, $(-2, -3)$

2. Reflection in the $x$-axis then the $y$-axis equals

   ANSWER A rotation of $180°$ about $O$

3. The translation that undoes $3$ right and $2$ down

   ANSWER $3$ left and $2$ up

4. The rotation that undoes $90°$ anticlockwise

   ANSWER $90°$ clockwise

5. A point on the mirror line maps to

   ANSWER Itself

6. Two rotations of $90°$ anticlockwise about $O$ make

   ANSWER A rotation of $180°$

7. Which transformation leaves no point fixed?

   ANSWER A translation

## 07 Homework

### Homework — 5 tasks

Describe every transformation fully, and label the image $A'$, $B'$, $C'$.

1. Plot $A(1, 2)$, $B(4, 2)$, $C(1, 4)$ and reflect the triangle in the $y$-axis.
2. Reflect the original triangle in the $x$-axis and write the three image points.
3. Rotate the original triangle $90°$ clockwise about the origin.
4. Translate the original triangle $4$ left and $3$ down.
5. State which of your four images have the reversed orientation, and explain why.